

Assimilation Across Accelerator Classifier Architectures: A Cross-Family Metrology of Directional Failure in Real-Time Learned Selection (EA-SEI-ACRB-01 v1.0)

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Abstract

Defines the Accelerator Classifier Retention Battery (ACRB): a pre-registered cross-family measurement framework for determining whether directional assimilation is specific to reconstruction autoencoders or persists across materially different learned-selection architectures. The battery compares architectures at matched own-background operating points, follows models through deployment transformations, and measures not only global discrimination but the topology and overlap of their miss regions.

The design is deliberately symmetric in outcome: either broad persistence or architecture-specific remediation is a substantive result. If normalized autoencoders or another architecture remove directional asymmetry across the pre-registered pair set, ACRB functions as a validation instrument for that remedy; if assimilation persists across reconstruction, latent, density, flow, distilled, and open-set families, the evidentiary burden shifts from a single-model defect toward a selection-system phenomenon. Claim boundaries are stated explicitly: assimilation is operational (a defined withheld class falling on the ordinary side of a calibrated threshold), AUC and score-order inversion and threshold-level retention are distinct estimands that must not be collapsed, public benchmark results do not establish deployed experiment efficiencies, and architectural diversity does not imply diversity of failure — miss overlap is an empirical quantity, measured as the RII family. Teacher-to-student blind-spot inheritance and float-to-firmware retention drift are treated as first-class deployment questions.

Body

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Real-Time Learned Selection (EA-SEI-ACRB-01 v1.0)

Description

Defines the Accelerator Classifier Retention Battery (ACRB): a pre-registered cross-family measurement framework for determining whether directional assimilation is specific to reconstruction autoencoders or persists across materially different learned-selection architectures. The battery compares architectures at matched own-background operating points, follows models through deployment transformations, and measures not only global discrimination but the topology and overlap of their miss regions.

The design is deliberately symmetric in outcome: either broad persistence or architecture-specific remediation is a substantive result. If normalized autoencoders or another architecture remove directional asymmetry across the pre-registered pair set, ACRB functions as a validation instrument for that remedy; if assimilation persists across reconstruction, latent, density, flow, distilled, and open-set families, the evidentiary burden shifts from a single-model defect toward a selection-system phenomenon. Claim boundaries are stated explicitly: assimilation is operational (a defined withheld class falling on the ordinary side of a calibrated threshold), AUC and score-order inversion and threshold-level retention are distinct estimands that must not be collapsed, public benchmark results do not establish deployed experiment efficiencies, and architectural diversity does not imply diversity of failure — miss overlap is an empirical quantity, measured as the RII family. Teacher-to-student blind-spot inheritance and float-to-firmware retention drift are treated as first-class deployment questions.

Methodology

Comparative and specification work built on published accelerator-trigger documentation (CMS Level-1 Data Scouting, Global Trigger, AXOL1TL, CICADA; ATLAS GELATO; LHCb, CBM, Belle II) and on the anomaly-detection methods literature; claim boundaries stated explicitly in the paper; SPXI packet subordinated to the prose. Terminology reconciled at mint: the irreversibility frontier supersedes the irreversibility locus of #1450, and cross-model miss overlap is unified with the Representational Independence Index family of #1450.

Falsification Conditions

The persistence claim fails if, across the pre-registered pair set at matched operating points, symmetric-by-design architectures (NAE/WNAE) or any other family show no directional score deficit — in which case ACRB records the remedy rather than the pathology. The redundancy claim fails if measured Δ_{miss} across paired architectures is consistently near zero or negative. The Low-Complexity Blind-Spot Hypothesis fails if classes below the training distribution on a measured representation-complexity axis show no systematic score deficit relative to comparably separated classes above it.

Files

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Abstract

Unsupervised anomaly detection is increasingly proposed or deployed as a means of extending collider sensitivity beyond explicitly targeted signal models. Yet the operational claim that a learned selector is “signal-agnostic” or “model-independent” is stronger than the claim that it performs well on a finite set of benchmark signals. A model can be agnostic with respect to signal labels during training while remaining highly selective with respect to the geometry, complexity, and representation of departures from its learned reference distribution.

A demonstration-scale battery motivating the present study reproduces a known directional asymmetry in collider autoencoders and observes related behavior across reconstruction, latent-space, and density-score families: when the direction of the normal/anomalous pair is reversed, the nominally anomalous class can receive *more ordinary* scores than the class on which the model was trained. The purpose of the present paper is not to generalize that result by assertion, but to convert it into a cross-architecture measurement program.

We introduce the **Accelerator Classifier Retention Battery (ACRB)**, a pre-registered framework for measuring directional assimilation across reconstruction autoencoders, encoder-side variational scores, explicit density estimators, normalizing flows, normalized autoencoders, distilled hardware triggers, and supervised real-time classifiers. The primary quantities are defined at matched operating points rather than by aggregate discrimination alone: partner-class assimilation, bidirectional retention gaps, score-order inversion, teacher-to-student blind-spot inheritance, float-to-firmware retention drift, and pairwise miss overlap. The design separates three questions often conflated in accelerator anomaly-detection studies: whether a model discriminates a chosen benchmark; whether its anomaly score is directionally ordered under reversal of the reference class; and whether its deployed threshold retains structurally distinct events at the rate budget under which the instrument actually operates.

The central hypothesis is deliberately architecture-neutral. The relevant common structure may not be the autoencoder. It may be the placement of a learned representation and scalar score upstream of irreversible retention. If directional assimilation survives changes of model family, representation, and hardware transformation, it should be treated as a selection-system property requiring retention metrology. If it disappears under specific normalized, plural, or open-set architectures, the same battery becomes a validation instrument for remedies. Either result is scientifically useful.

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